Climate and Tourism in Europe

Are Cooler Destinations Becoming More Attractive?

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Abstract

Tourism in Europe exhibits strong spatial and temporal variation, often associated with environmental conditions. This paper investigates the relationship between climate variables and tourism activity across European regions using panel data at the NUTS2 level over the period 2010–2023. Combining Eurostat tourism statistics with climate data derived from the Open-Meteo API, the study examines how temperature and precipitation are associated with tourism flows across space and time.

The analysis adopts a relational approach and relies on panel data methods, including fixed effects specifications, complemented by machine learning techniques to explore non-linear patterns. The results indicate that temperature has a positive and statistically significant within-region effect on tourism, while precipitation does not exhibit a robust relationship. Evidence of non-linearity emerges in the econometric specification, suggesting an inverted U-shaped relationship, although this pattern is less pronounced in the machine learning results.

Across all models, economic factors—particularly GDP—consistently emerge as the primary drivers of tourism activity. Overall, the findings suggest that climate influences tourism at the margin, but does not fundamentally reshape tourism patterns once regional heterogeneity is accounted for. In particular, the analysis does not provide strong evidence of a systematic shift in tourism demand toward cooler regions over the period.

These results highlight both the relevance and the limitations of climate variables in explaining tourism dynamics, and underscore the importance of structural economic conditions in shaping regional tourism activity.

Keywords: tourism, climate, panel data, temperature, machine learning, Europe

Word count: 6,500

1 Introduction

Tourism represents a major component of economic activity across Europe, contributing significantly to regional development, employment, and international mobility. However, tourism flows are highly uneven across both space and time, with pronounced seasonal peaks and strong regional disparities.

Climate conditions are often considered a key factor shaping tourism patterns. Temperature and precipitation influence travel decisions, destination attractiveness, and the feasibility of outdoor activities. In particular, warmer and drier conditions are commonly associated with increased tourism demand in coastal and leisure-oriented regions.

At the same time, existing empirical evidence remains inconclusive regarding the strength and form of climate effects. While climate is frequently identified as an important determinant, its role is difficult to isolate due to strong correlations with seasonal cycles and structural regional characteristics such as economic development and infrastructure.

This raises a broader question: to what extent do climate conditions independently structure tourism activity, and can they lead to systematic shifts in tourism demand in a context of rising temperatures?

This study adopts a computational social science perspective to address this issue using large-scale regional data. By combining panel data methods with machine learning approaches, the analysis aims to characterise both linear and non-linear climate-tourism relationships while explicitly accounting for regional heterogeneity.

This study contributes to the literature in three main ways. First, it provides a harmonised regional panel dataset at the NUTS2 level covering a large number of European regions over more than a decade, allowing for a consistent comparison of climate-tourism relationships across space and time.

Second, it shows that once regional heterogeneity is accounted for, climate variables—particularly temperature—have a statistically significant but economically limited effect on tourism activity, while structural economic factors remain the dominant drivers.

Third, by combining panel econometric models with machine learning approaches, the study highlights how different modeling frameworks capture distinct aspects of the data, with flexible models emphasising cross-sectional variation and econometric specifications isolating within-region effects.

Taken together, these findings provide a nuanced perspective on the role of climate in shaping tourism dynamics, suggesting that climate influences tourism primarily at the margin rather than as a central determinant of regional tourism patterns.

Rather than attempting to establish causal effects, the paper focuses on identifying robust statistical patterns and assessing the relative importance of climate relative to structural economic factors.

All data processing and analysis steps are fully reproducible and documented in Section 9.

2 Background and Related Work

2.1 Climate Change and Tourism Dynamics

Climate change constitutes one of the most significant structural transformations affecting socio-economic systems worldwide. In Europe, rising temperatures, increasing frequency of extreme weather events, and shifting precipitation patterns are expected to alter the spatial and temporal distribution of economic activities (IPCC, 2021). As a climate-sensitive sector, tourism is particularly exposed to these changes, as environmental conditions directly influence destination attractiveness, travel decisions, and the feasibility of tourism-related activities.

Recent contributions highlight that climate change is not only a long-term environmental issue but also a key economic factor shaping regional competitiveness and sectoral dynamics. In the tourism sector, these changes may affect both the supply side, through infrastructure and environmental constraints, and the demand side, through changing preferences and travel patterns (Scott et al., 2016). As a result, understanding the relationship between climate conditions and tourism activity has become an increasingly important research topic.

2.2 Climate as a Determinant of Tourism Demand

A substantial body of research has examined the relationship between climate conditions and tourism demand. Early contributions focused on the role of weather and climate as key determinants of destination attractiveness, particularly for leisure tourism (Maddison, 2001; Lise and Tol, 2002). These studies generally find that tourists exhibit strong preferences for specific climatic conditions, typically favouring warm and dry environments.

Building on this framework, Hamilton et al. (2005) and Bigano et al. (2006) developed macro-level models of international tourism flows, showing that climate variables significantly shape global tourism patterns. These approaches emphasise the importance of climatic conditions in determining the spatial allocation of tourism demand at the global scale.

More recent work has expanded this perspective by incorporating climate change projections and exploring their long-term implications. Scott et al. (2012, 2019) and Gössling et al. (2012) argue that rising temperatures may lead to a geographical redistribution of tourism demand, potentially reducing the attractiveness of already warm regions while benefiting cooler destinations. Empirical analyses also suggest that the relationship between temperature and tourism is non-linear, with diminishing or even negative effects at high temperature levels (Rutty and Scott, 2010; Falk, 2014). These findings indicate that climate effects are complex and may vary across contexts and temperature ranges.

2.3 Climate vs Weather: Conceptual Distinction

An important conceptual distinction in the literature concerns the difference between weather and climate. Weather refers to short-term atmospheric conditions, while climate captures long-term statistical patterns. Tourism decisions may respond differently to these two dimensions. Short-term weather shocks can influence immediate travel behaviour, whereas long-term climate conditions shape destination attractiveness and structural tourism patterns (Dell et al., 2014).

From an empirical perspective, this distinction has important implications. Many studies rely on high-frequency weather data to capture short-term effects, while others focus on long-term climate averages. However, these approaches may capture different underlying mechanisms. In particular, climate variables are often correlated with geographic characteristics and persistent regional attributes, making it difficult to disentangle their independent effect from structural factors (Dell et al., 2014; Burke et al., 2015).

In this study, the use of annual aggregated climate data places the analysis closer to a climatic perspective, focusing on medium-term conditions rather than short-term weather fluctuations. This choice reflects both data availability constraints and the objective of analysing broad regional patterns across Europe.

2.4 Heterogeneity of Climate Effects

Another important dimension concerns the heterogeneity of climate effects across different types of tourism and regions. Tourism demand is not homogeneous and may respond differently to climatic conditions depending on the nature of destinations (Falk, 2014; Scott et al., 2019). Coastal regions, for instance, may benefit from higher temperatures up to a certain threshold, while urban tourism may be less sensitive to climatic variations. Similarly, mountain regions may exhibit distinct seasonal dynamics, where temperature increases can have both positive and negative effects depending on the context.

This heterogeneity implies that aggregate estimates may mask substantial variation in climate-tourism relationships. It also motivates the use of regional-level data, as adopted in this study, to better capture spatial differences in responses to climate conditions. By focusing on NUTS2 regions, the analysis allows for a more granular examination of these heterogeneous effects across Europe.

2.5 Empirical and Methodological Limitations

Despite these advances, several limitations persist in the empirical literature. First, many studies rely on aggregated national data or focus on specific destinations, limiting the ability to capture regional heterogeneity within countries (Falk, 2014). Yet tourism systems are inherently localised, and responses to climate conditions may vary substantially across regions depending on infrastructure, economic structure, and tourism specialisation.

Second, a large share of the literature relies on relatively low-frequency data (annual or seasonal aggregates), which may obscure important intra-annual dynamics (Baum and Lundtorp, 2001). Tourism seasonality is indeed a central feature of the phenomenon (Butler, 1994), and higher-frequency data would in principle allow for a more precise analysis of short-term co-movements between climate conditions and tourism activity.

However, the use of high-frequency data in a comparative regional setting raises substantial challenges. In particular, harmonised datasets combining tourism and climate indicators at fine temporal resolution remain limited in terms of spatial coverage and cross-country comparability. As a result, empirical analyses at a broader geographical scale often rely on aggregated data to ensure consistency across regions.

From a methodological perspective, the use of annual data is not only a constraint but also reflects a deliberate trade-off. By focusing on lower-frequency variation, the analysis abstracts from short-term weather fluctuations and seasonal noise, allowing for a clearer identification of medium-term relationships between climate conditions and tourism activity. This approach is particularly suited to panel data frameworks, where the objective is to exploit both cross-sectional and temporal variation while controlling for unobserved heterogeneity.

Third, observational studies face well-known challenges related to identification. Climate variables are strongly correlated with seasonal cycles and may co-evolve with other unobserved factors influencing tourism demand, such as economic conditions, infrastructure development, or policy interventions (Falk, 2014). In this context, the use of annual aggregates can help mitigate some of these issues by smoothing high-frequency variability and reducing the influence of short-term shocks, although it does not fully resolve identification concerns. As a result, disentangling the independent contribution of climate remains methodologically challenging.

2.6 Endogeneity and Identification Challenges

A further methodological issue relates to potential endogeneity in the relationship between climate and tourism. While climate conditions influence tourism demand, tourism development itself may affect regional characteristics such as infrastructure, accessibility, and economic activity. These factors may, in turn, be correlated with climate, leading to biased estimates if not properly accounted for. Similar concerns have been highlighted in the empirical tourism literature, where omitted variables and reverse causality complicate inference (Falk, 2014).

In addition, climate variables are often correlated with seasonal patterns and other time-varying factors, making it difficult to isolate their independent effect. This issue is closely related to the broader identification problem in observational data, where explanatory variables co-vary with unobserved determinants of the outcome (Angrist and Pischke, 2009). In the context of climate-tourism relationships, this is particularly relevant, as climatic conditions are intertwined with geographic, economic, and institutional factors.

These challenges highlight the limits of causal interpretation in this setting. As a result, recent contributions emphasise the importance of transparent empirical strategies and careful interpretation of results, especially when relying on non-experimental data (Salganik, 2017).

2.7 Research Gap and Contribution

Taken together, these limitations point to a more specific gap in the literature. While existing studies have established that climate conditions are associated with tourism activity, the joint analysis of climate and tourism variables at a comparable regional scale remains constrained by data availability and comparability issues. In particular, harmonised datasets combining economic and climate indicators at the regional level are relatively scarce, limiting the scope for large-scale comparative analysis.

Beyond these empirical constraints, fewer contributions combine harmonised regional data with both econometric and machine learning approaches within a unified empirical framework. This limitation restricts the ability to compare linear and non-linear representations of climate-tourism relationships.

More fundamentally, this gap reflects a broader methodological divide in statistical modeling between the "data modeling culture" and the "algorithmic modeling culture" (Breiman, 2001). The former relies on predefined stochastic models to characterise relationships, while the latter emphasises predictive performance using flexible, data-driven approaches. In contexts such as climate-tourism interactions, where underlying mechanisms are complex and potentially non-linear, relying exclusively on parametric specifications may lead to model misspecification and incomplete representations of the data-generating process (Breiman, 2001).

Given the observational nature of most available data, identifying causal effects is inherently challenging, as climate variables co-vary with seasonal patterns and broader socio-economic dynamics. As a result, there is a need for empirical approaches that prioritise transparent documentation of statistical relationships rather than causal interpretation.

The present study addresses these gaps by constructing a harmonised panel dataset at the NUTS2 level covering European regions over the period 2010–2023. By combining tourism indicators from Eurostat with climate data derived from ERA5 reanalysis (Copernicus Climate Data Store), the analysis enables a systematic comparison of regional patterns within a consistent framework.

Methodologically, the paper adopts a panel data approach to exploit both cross-sectional and temporal variation in the data. This design allows for the identification of robust associations between climate conditions and tourism activity while controlling for unobserved regional heterogeneity and common temporal trends.

More specifically, this study positions itself at the intersection of empirical tourism economics and computational social science. While the former provides a structured econometric framework to analyse climate-tourism relationships, the latter allows for the integration of large-scale heterogeneous data and flexible modelling approaches. This dual perspective enables a more comprehensive understanding of tourism dynamics, combining interpretability with data-driven exploration. In doing so, the paper contributes to the literature by providing both a harmonised empirical dataset and a methodological framework that allows for the comparison of alternative modeling paradigms.

3 Research Problem

This study addresses the following research question:

> What is the relationship between climate conditions and tourism activity across European regions over time?

The analysis adopts a relational rather than causal perspective, focusing on the identification of statistical associations between climate variables and tourism flows. Given the observational nature of the data and the absence of a clear identification strategy, the objective is not to estimate causal effects but to provide a robust empirical characterisation of climate-tourism relationships.

To address this question, the study adopts a longitudinal panel design, exploiting both cross-sectional variation across regions and temporal variation over the period 2010–2023. The use of annual data, while limiting the analysis of intra-annual dynamics, ensures consistency and comparability across regions and allows for the identification of medium-term patterns in tourism activity.

Identification Challenges

From an empirical perspective, several identification challenges arise when analysing the relationship between climate conditions and tourism activity. First, omitted variable bias may occur if relevant factors such as infrastructure, accessibility, or tourism specialisation are not fully captured in the model. These factors are likely to be correlated with both climate conditions and tourism demand, potentially biasing estimated relationships.

Second, simultaneity and reverse causality cannot be entirely ruled out. While climate conditions plausibly influence tourism activity, tourism development may also affect regional characteristics, such as economic activity or infrastructure, which are themselves correlated with climatic conditions.

Third, climate variables are strongly correlated with seasonal patterns and broader time-varying dynamics. This creates a form of seasonal confounding, where variations in tourism activity may reflect underlying seasonal cycles rather than independent climate effects.

More generally, these issues reflect the broader identification problem in observational data, where explanatory variables co-vary with unobserved determinants of the outcome (Angrist and Pischke, 2009). As a result, the analysis focuses on identifying robust statistical relationships rather than causal effects, in line with recent approaches in computational social science (Salganik, 2017).

Based on the existing literature, two empirical expectations guide the analysis:

H1: Tourism activity is positively associated with higher temperatures and negatively associated with precipitation.

H2: The strength of this relationship varies across regions, reflecting differences in tourism specialisation, environmental conditions, and regional characteristics.

These hypotheses are formulated in relational terms and are tested within a panel data framework that allows for the control of unobserved regional heterogeneity and common temporal trends.

4 Data

The data and code used in this study are publicly available in a GitHub repository (see Section 9). The repository provides a fully reproducible version of the analysis, including all scripts, processed data, and presentation materials.

The analysis combines multiple data sources to construct a harmonised regional panel dataset integrating economic, demographic, and environmental information at the NUTS2 level.

Tourism data are obtained from Eurostat using the dataset `tour_occ_nin2`, which provides information on nights spent in accommodation establishments. The raw dataset is a large multi-dimensional panel covering multiple units, economic sectors, and regions. After filtering for total nights (unit = NR) and accommodation services (NACE I551), and restricting the analysis to NUTS2 regions, the resulting dataset contains 4,365 region-year observations.

Gross domestic product (GDP) data are retrieved from Eurostat (`nama_10r_2gdp`) and filtered to retain total GDP (B1GQ) expressed in million euros. Population data are obtained from Eurostat (`demo_r_pjanaggr3`) and filtered for total population. After reshaping and harmonisation, these datasets contain 7,498 and 9,509 observations respectively.

Climate data are derived from the ERA5 reanalysis dataset (Copernicus Climate Data Store), accessed via the Open-Meteo API. They include temperature and precipitation variables at a monthly spatial grid level. These data are aggregated to annual values and spatially matched to NUTS2 regions using geographic coordinates and polygon boundaries. This spatial aggregation step is necessary to translate gridded climate information into administratively comparable units.

Temperature is computed as annual averages, while precipitation is aggregated as total annual accumulation. This aggregation reflects a trade-off between temporal precision and cross-regional comparability, and is consistent with the focus on medium-term climate conditions.

The integration of these heterogeneous data sources requires substantial harmonisation, including the alignment of regional identifiers, the reconciliation of different spatial and temporal resolutions, and the treatment of missing or inconsistent observations. The intersection of the economic and demographic datasets yields 293 regions prior to the inclusion of climate data.

After merging all data sources and excluding observations with missing climate information, the final dataset covers 273 European regions across 34 countries over the period 2010–2023, resulting in a total of 3,274 observations.

The dataset constitutes an unbalanced panel combining cross-sectional and temporal variation, with a coverage ratio of approximately 85.7% of all possible region-year observations. This unbalanced structure reflects differences in data availability across regions and sources.

By combining multiple large-scale datasets into a unified empirical framework, the resulting panel provides a consistent and comparable basis for analysing climate-tourism relationships across Europe, while integrating economic, demographic, and environmental dimensions.

5 Operationalisation

The dependent variable in this study is tourism activity, measured as the annual number of nights spent in accommodation establishments at the regional level. Given the strong right-skewness of this variable, a logarithmic transformation is applied to reduce heteroskedasticity and allow for interpretation in relative terms.

The key explanatory variables capture climatic conditions. Temperature is measured as the annual average (in degrees Celsius), while precipitation corresponds to total annual accumulation (in millimetres). Both variables are treated as continuous predictors and reflect the overall climatic environment within each region-year.

Additional control variables include gross domestic product (GDP) and population, which capture economic development and demographic scale. These controls account for structural differences in tourism capacity and demand across regions.

All variables are defined at the NUTS2 regional level and aggregated on an annual basis. The use of annual data is motivated by the need to ensure consistency and comparability across heterogeneous data sources, particularly when combining economic, demographic, and spatially aggregated climate data. While higher-frequency data could capture intra-annual dynamics, such data remain difficult to harmonise at the regional level in a comparative European setting. Annual aggregation therefore provides a robust framework for analysing medium-term patterns across regions and over time.

From an econometric perspective, the logarithmic transformation implies that estimated coefficients can be interpreted as semi-elasticities, relating changes in climate variables to percentage variations in tourism activity.

6 Analytical Approach

The empirical strategy combines panel data regression techniques with complementary machine learning methods to analyse the relationship between climate conditions and tourism activity.

6.1 Baseline and Panel Specifications

The baseline specification is estimated using ordinary least squares with time fixed effects:

$$\log(Tourism_{r,t}) = \beta_1 Temperature_{r,t} + \beta_2 Precipitation_{r,t} + \beta_3 \log(GDP_{r,t}) + \beta_4 \log(Population_{r,t}) + \lambda_t + \epsilon_{r,t}$$

This specification captures both cross-sectional and temporal variation, while controlling for common time shocks affecting all regions.

To account for unobserved heterogeneity across regions, a fixed effects panel model is estimated:

$$\log(Tourism_{r,t}) = \beta_1 Temperature_{r,t} + \beta_2 Precipitation_{r,t} + \alpha_r + \lambda_t + \epsilon_{r,t}$$

where α_r captures region-specific effects and λ_t common time effects. Standard errors are clustered at the regional level to account for within-region correlation over time.

This specification exploits within-region variation over time, allowing the identification of the effect of climate fluctuations relative to region-specific averages.

6.2 Non-linear Specification

To account for potential non-linear effects of temperature, an alternative specification includes a quadratic term:

$$\log(Tourism_{r,t}) = \beta_1 Temp^*_{r,t} + \beta_2 (Temp^*_{r,t})^2 + \beta_3 Precipitation_{r,t} + \alpha_r + \lambda_t + \epsilon_{r,t}$$

where $Temp^*_{r,t}$ denotes temperature anomalies, defined as deviations from the regional mean.

This transformation removes cross-sectional differences in climate levels and focuses on deviations from typical local conditions. The quadratic specification allows identifying potential threshold effects in the relationship between temperature and tourism activity.

The implied optimal temperature is derived as:

$$T^* = -\frac{\beta_1}{2\beta_2}$$

6.3 Machine Learning Approach

To complement the linear models, machine learning methods are employed as exploratory tools. A decision tree model is estimated to capture non-linearities and interaction effects in an interpretable manner. In addition, a random forest model is used to assess variable importance and predictive performance.

The models are trained on the same set of variables as the econometric specifications, ensuring comparability across approaches. Partial dependence plots are used to visualise the marginal effects of key variables, providing further insights into non-linear relationships.

These methods are used for descriptive and exploratory purposes only and do not aim to provide causal inference.

6.4 Interpretation Framework

Across all specifications, the analysis focuses on identifying statistical relationships rather than causal effects, in line with the observational nature of the data. The combination of econometric and machine learning approaches allows for a more comprehensive characterisation of climate-tourism dynamics across European regions, balancing interpretability and flexibility.

7 Results

7.1 Descriptive Patterns

Figure 1 presents the distribution of tourism activity across regions in logarithmic scale. The transformation produces a relatively symmetric distribution, indicating that the log specification effectively reduces skewness and stabilises variance.

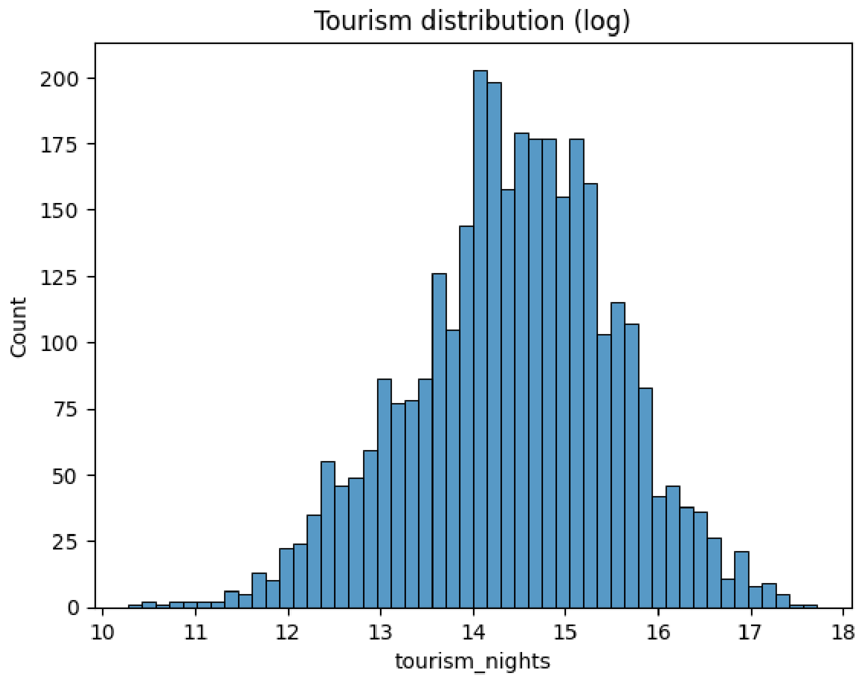

Figure 1: Distribution of tourism activity (log scale)

Figures 2 and 3 display the spatial distribution of temperature and tourism activity across European regions. A clear north–south gradient is observed for temperature, with warmer conditions in Southern Europe. Tourism activity is also relatively higher in many of these regions, suggesting a positive spatial association between climate conditions and tourism.

However, this relationship is not uniform across space. Some regions with similar climatic conditions exhibit different levels of tourism activity, indicating that climate alone does not fully explain tourism patterns. This highlights the role of additional structural factors, such as economic development, infrastructure, and tourism specialisation.

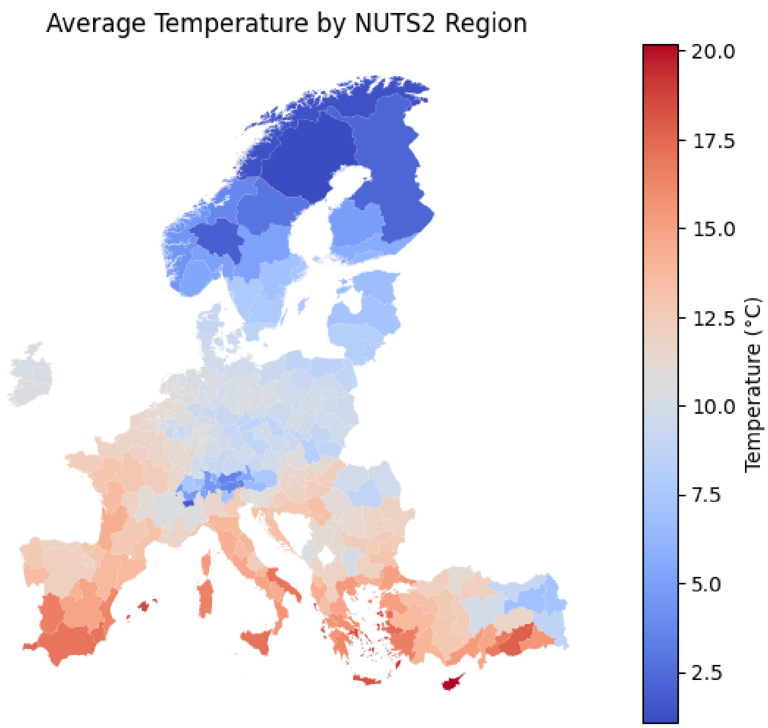

Figure 2: Average temperature by NUTS2 region

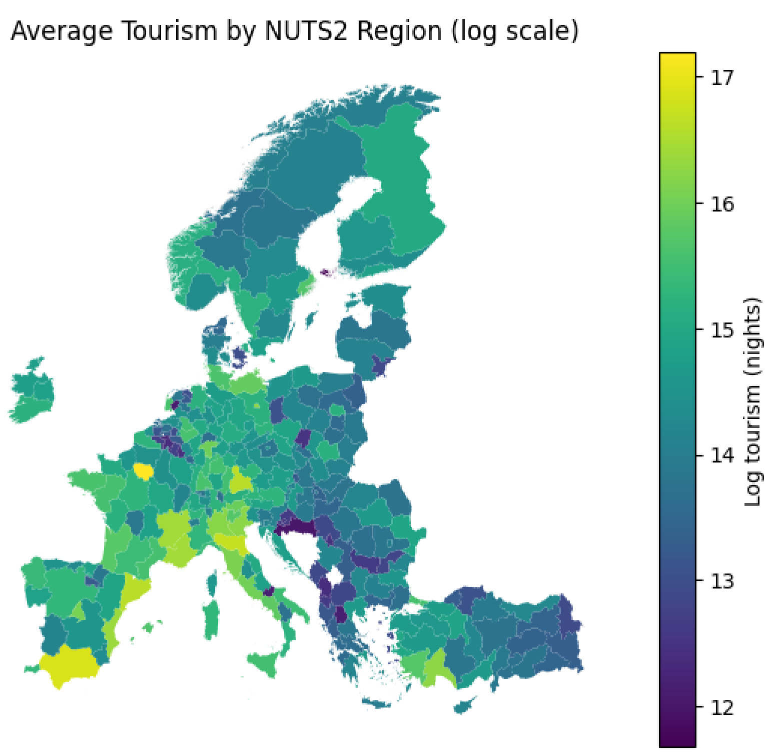

Figure 3: Average tourism activity by NUTS2 region (log scale)

Figure 4 shows the evolution of average temperature over time. A clear upward trend is observed over the period 2010–2023, with an increase of approximately 2–2.5°C across the sample.

While this magnitude appears large compared to standard global warming estimates, it reflects aggregated regional averages and should not be interpreted as a direct measure of long-term climate change. Differences in spatial coverage, aggregation methods, and data structure may contribute to this variation.

Nevertheless, the figure provides clear evidence of a rising temperature trend in Europe over the study period. This pattern supports the relevance of analysing the relationship between climate conditions and tourism activity, as it indicates that climatic conditions are evolving in a way that may affect regional attractiveness.

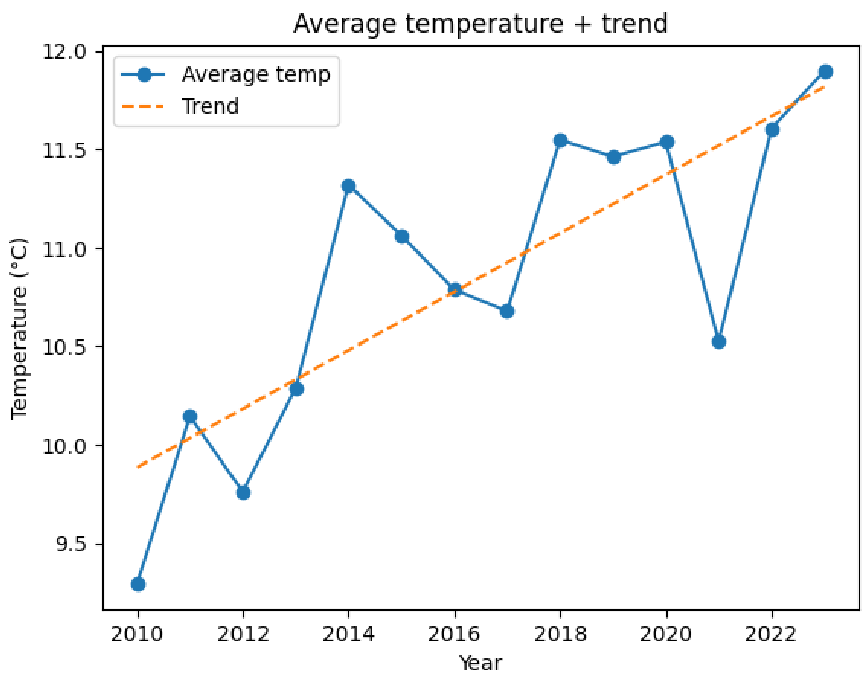

Figure 4: Average temperature over time

7.2 Baseline and Panel Estimates

Table 1 reports the results from the OLS and fixed effects panel regressions. The specifications differ in the type of variation they exploit, allowing for a comparison between cross-sectional patterns and within-region dynamics over time. In the OLS model, GDP and population are included as control variables capturing structural differences in economic size and tourism capacity across regions.

The OLS results indicate that economic variables are the main drivers of tourism activity, with GDP exhibiting a strong and statistically significant association ($\beta = 0.620$, $p < 0.01$). Population also has a positive effect ($\beta = 0.174$, $p < 0.1$), while the coefficient on temperature is relatively small and only marginally significant ($\beta = 0.034$, $p \approx 0.05$). Precipitation appears positive and statistically significant in this cross-sectional specification ($\beta = 6.13 \times 10^{-5}$, $p < 0.01$).

Taken together, these results primarily reflect differences across regions rather than changes within regions over time. In particular, richer regions tend to exhibit higher tourism activity, and these structural differences dominate the cross-sectional relationships observed in the data.

In the fixed effects framework, region-specific characteristics are absorbed by entity fixed effects. This means that the estimation focuses exclusively on variation within each region over time, effectively controlling for all time-invariant characteristics such as geography, baseline climate, or long-term economic structure. As a result, variables such as GDP and population, which evolve slowly and primarily capture cross-sectional differences, provide limited additional information and are excluded from these specifications.

To better isolate short-term climate variation, temperature is expressed as an anomaly, defined as the deviation from the region-specific mean:

$$Temp^{*}_{r,t} = Temp_{r,t} - T\bar{e}mp_{r}$$

This transformation removes persistent cross-sectional differences in climate levels and allows the analysis to focus on deviations from typical local conditions.

Once region fixed effects are introduced, the magnitude and interpretation of climate variables change substantially. The effect of temperature becomes larger and highly significant when expressed in terms of deviations from the regional mean ($\beta = 0.171$, $p < 0.01$).

In practical terms, this implies that a one-degree increase relative to usual climatic conditions within a given region is associated with an increase of approximately 17% in tourism activity. This effect is economically meaningful in relative terms, as it captures how short-term variations in climate conditions may influence tourism demand. At the same time, it should be interpreted as a temporary deviation rather than a long-run structural shift in tourism patterns.

In contrast, precipitation loses statistical significance ($\beta = 1.50 \times 10^{-5}$, $p > 0.5$), suggesting that its apparent effect in the OLS specification is largely driven by persistent differences across regions rather than meaningful variation over time within regions.

However, the relationship between temperature and tourism is unlikely to be purely linear. To account for potential non-linear effects, a quadratic term is introduced. The results reveal a statistically significant inverted U-shaped relationship between temperature and tourism. The linear term remains positive ($\beta = 0.133$, $p < 0.01$), while the squared term is negative and significant ($\beta = -0.119$, $p < 0.01$), indicating diminishing returns to temperature.

This implies that tourism activity increases with temperature up to a certain point, after which additional increases in temperature are associated with a decline in tourism activity. The implied optimal temperature lies around 11–12°C.

Formally, this turning point is obtained as:

$$T^* = -\frac{\beta_1}{2\beta_2}$$

where β_1 and β_2 denote the coefficients on the linear and squared temperature terms. This value should be interpreted as a statistical optimum within the observed data rather than as a universal behavioural threshold.

Despite these statistically significant effects, the overall explanatory power of climate variables remains limited. The within R^2 of the fixed effects models is relatively low (0.133 and 0.169), indicating that climate fluctuations explain only a small share of intra-regional variation in tourism over time.

This result highlights the dominant role of structural and economic factors in shaping tourism patterns. While climate conditions do matter, their influence remains secondary compared to persistent differences in economic capacity across regions. In particular, the magnitude of the temperature effect remains modest relative to the cross-sectional variation captured by GDP, reinforcing the conclusion that economic development is the primary determinant of tourism activity in the European context.

Table 1: Climate and tourism: panel regression results

	(1) OLS	(2) FE	(3) FE Non-linear
Temperature	0.034*		
	(0.018)		
Temperature anomaly		0.171***	0.133***
		(0.040)	(0.032)
Temperature anomaly^2			-0.119***
			(0.028)
Precipitation	6.13e-05***	1.50e-05	-1.24e-06
	(1.82e-05)	(2.27e-05)	(2.51e-05)
log(GDP)	0.620***		
	(0.058)		
log(Population)	0.174*		
	(0.086)		
Intercept	4.563***	14.412***	14.480***
	(0.863)	(0.025)	(0.031)
Observations	3274	3274	3274
Regions	273	272	272
Year FE	Yes	Yes	Yes
Region FE	No	Yes	Yes
R^2	0.528		
Within R^2		0.133	0.169

Standard errors in parentheses, clustered at the regional (entity) level.
*** p<0.01, ** p<0.05, * p<0.1
Temperature anomaly defined as deviation from the regional mean.
Year fixed effects included (base year: 2010).

7.3 Non-linear Climate Effects

Figure 5 presents the estimated non-linear relationship between temperature and tourism activity. The results reveal a clear inverted U-shaped relationship: tourism increases with temperature up to a threshold, after which higher temperatures are associated with a decline in tourism activity.

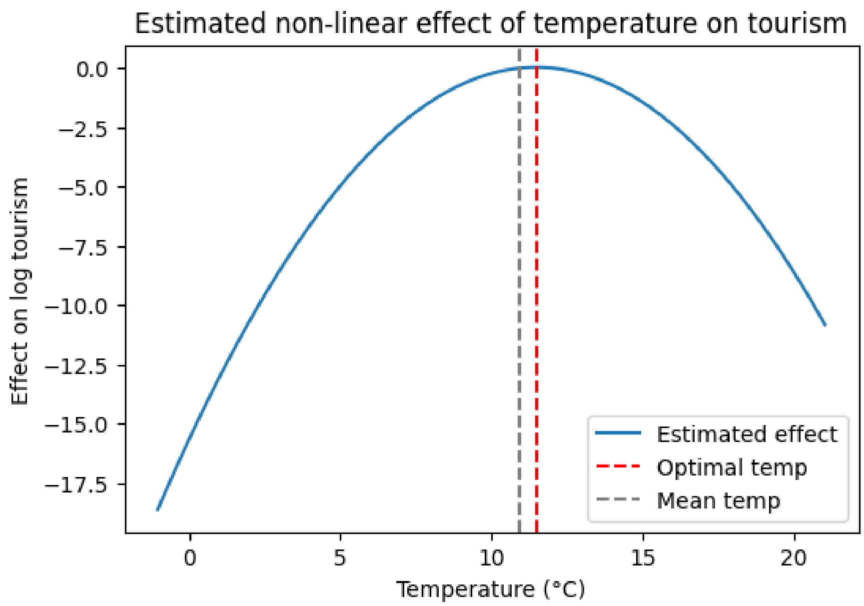

Figure 5: Estimated non-linear effect of temperature on tourism

The estimated optimal temperature is approximately 11–12°C. This value should be interpreted cautiously, as it reflects a statistical optimum within the panel rather than an absolute preference level.

7.4 Machine Learning Evidence

While the econometric models provide structured estimates, their explanatory power remains limited and may not fully capture complex non-linearities. Machine learning methods are therefore used as exploratory tools to provide a flexible, data-driven characterization of climate-tourism relationships and to detect potential threshold or extreme effects. Tree-based methods are preferred due to their interpretability and ease of implementation.

Figure 6 presents the segmentation obtained from the decision tree model. The structure is primarily driven by GDP thresholds, with the first split occurring at relatively high income levels. Regions with high GDP exhibit substantially higher tourism activity, while lower-income regions are further segmented into finer groups. Climate variables do not appear among the main splitting criteria, indicating that economic factors dominate the hierarchical structure of tourism patterns.

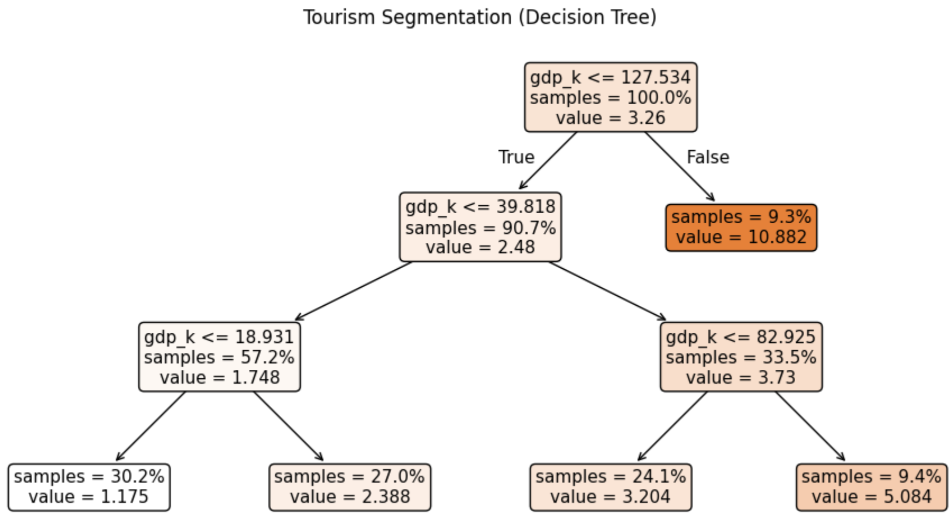

Figure 6: Tourism segmentation using a decision tree

Figure 7 shows the partial dependence plots derived from the random forest model, which represent the average marginal effect of each variable on predicted tourism while holding other variables constant. The relationship between temperature and tourism is weakly non-linear but remains predominantly increasing over most of the observed range, with only mild flattening at intermediate levels. This suggests that higher temperatures are generally associated with higher tourism activity, without a clear reversal at high values in the machine learning specification.

Precipitation exhibits a relatively flat and weak effect across most of its range, suggesting that variations in rainfall have limited influence on tourism activity at the annual level. In contrast, GDP displays a strong and monotonic relationship with tourism, with higher income levels consistently associated with higher predicted tourism activity. The steep increase observed at higher GDP levels indicates that economic capacity plays a central role in structuring tourism demand.

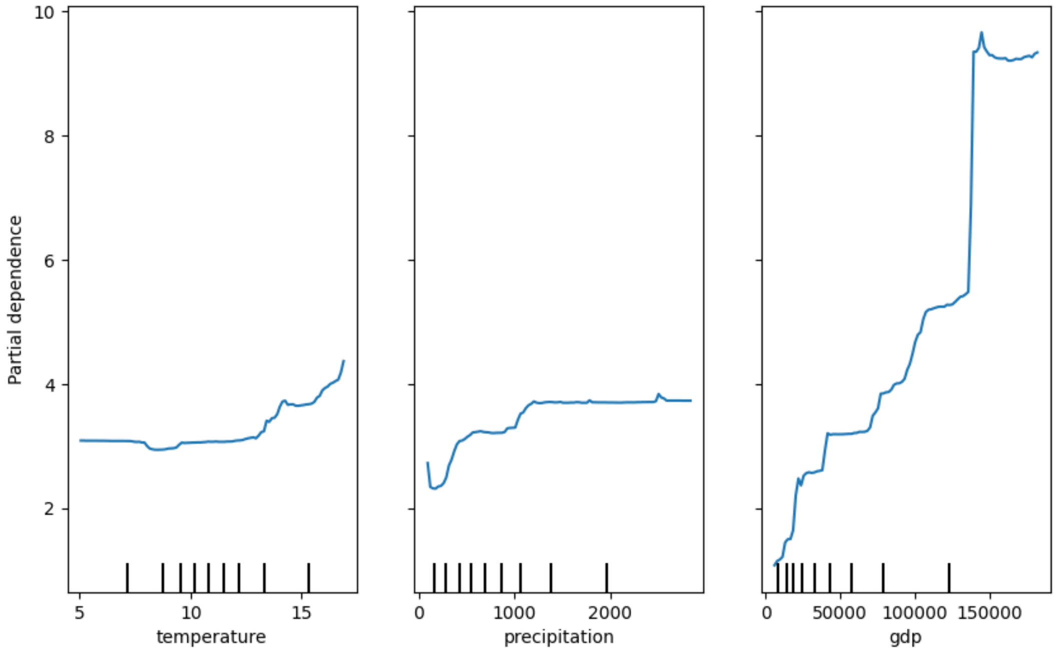

Figure 7: Partial dependence plots for key variables

The random forest model achieves a relatively high out-of-bag performance ($R^2 = 0.659$) with a root mean squared error of 2.44, indicating good predictive capacity. Variable importance measures further confirm the dominance of economic factors, with GDP contributing the most to predictive performance, followed by temperature and precipitation.

These results differ from the econometric findings in an important way. While the fixed effects models identify a non-linear, inverted U-shaped relationship between temperature and tourism based on within-region variation, the machine learning results suggest a more gradual and predominantly increasing relationship. One possible explanation is that machine learning models, by construction, are strongly influenced by dominant sources of variation in the data. In this case, cross-sectional differences—particularly in GDP—play a central role and may mask more subtle non-linear effects linked to short-term climate fluctuations.

Overall, these findings highlight that different modeling approaches capture distinct aspects of the data. Econometric models are better suited to isolating within-region dynamics, while machine learning models primarily reflect broader cross-sectional patterns. Taken together, the results reinforce the conclusion that while climate variables exhibit meaningful patterns, they remain secondary relative to structural economic determinants in explaining tourism activity.

7.5 Synthesis of Findings

Overall, the results indicate that climate conditions are associated with tourism activity, but their explanatory power remains limited once regional heterogeneity is accounted for. The fixed effects estimates show that short-term deviations in temperature within regions have a significant positive effect on tourism, with a one-degree increase relative to usual conditions associated with an increase of approximately 17% in tourism activity. In contrast, precipitation does not exhibit a robust impact once within-region variation is considered.

Evidence of non-linearity emerges in the econometric specification, suggesting the existence of an optimal temperature range. However, this pattern appears less pronounced in the machine learning results, where the relationship remains predominantly increasing with only mild non-linearities. This indicates that the strength and form of non-linear effects may be sensitive to model specification.

Across all approaches, economic factors, particularly GDP, consistently emerge as the primary drivers of tourism activity. Climate variables contribute to explaining variations in tourism, but their role remains secondary relative to structural economic determinants. In particular, the comparison between OLS and fixed effects models highlights that much of the observed variation in tourism is driven by persistent cross-sectional differences rather than short-term climate fluctuations.

Importantly, the differences between econometric and machine learning results highlight how model choice influences the interpretation of climate effects. Flexible models tend to emphasise dominant cross-sectional patterns, while econometric specifications isolate more subtle within-region dynamics.

Taken together, these findings suggest that climate influences tourism primarily at the margin, affecting short-term variations within regions, but does not fundamentally reshape tourism patterns when structural economic differences are taken into account. From a broader perspective, this implies that policies aimed at developing tourism are likely to be more effective when focused on economic capacity, infrastructure, and regional attractiveness rather than relying solely on climatic advantages.

8 Limitations

This study is subject to several limitations related to identification, data construction, and model specification.

Identification and omitted variables First, the use of observational data prevents causal interpretation. The estimated relationships reflect statistical associations and may be influenced by omitted variables such as infrastructure, policy, or regional tourism specialisation, which are not fully captured in the model.

Second, the analysis relies on a limited set of variables. While GDP and population capture key structural dimensions, other relevant factors, such as transport accessibility, tourism infrastructure, or institutional characteristics, are not included. This may lead to an incomplete representation of the determinants of tourism activity.

Temporal and spatial aggregation Third, climate variables are measured at an annual level, which implies a loss of information on seasonal dynamics. Tourism is inherently seasonal, and aggregating temperature and precipitation over the year may obscure short-term effects that are important for travel decisions. At the same time, this choice departs from traditional meteorological approaches, which typically rely on higher-frequency data. It reflects a trade-off between temporal precision and data availability, allowing for a consistent and comparable analysis across regions.

Fourth, climate data are spatially aggregated from gridded observations to NUTS2 regions. This reduces spatial precision and may mask local heterogeneity, particularly in geographically diverse regions. As a result, the constructed variables should be interpreted as regional averages rather than precise local conditions.

Measurement and interpretation Fifth, the construction of temperature anomalies and the use of aggregated values imply that the resulting variables are not directly interpretable as absolute climatic levels. Instead, they capture deviations from typical regional conditions, which limits their interpretation in terms of individual preferences or behavioural responses.

Sixth, the measure of tourism activity is based on nights spent in accommodation, which may not fully reflect tourism demand. It captures realised activity, which depends not only on climate conditions but also on supply-side constraints such as accommodation capacity.

Seventh, some empirical patterns may reflect data construction issues rather than underlying dynamics. For instance, the observed increase in average temperature over the period (Figure 4), which appears larger than standard climate change estimates, likely reflects aggregation effects, sample composition, or measurement differences.

Model dependence Finally, the results may be sensitive to model specification. While the econometric analysis suggests a non-linear relationship between temperature and tourism, the machine learning results indicate a more monotonic pattern. This discrepancy highlights that the form and magnitude of climate effects may depend on modelling choices.

Moreover, machine learning results may be influenced by the predominance of cross-sectional variation in the data. Variables such as GDP, which exhibit strong structural differences across regions, tend to dominate predictive performance, potentially masking more subtle within-region effects of climate variables.

Taken together, these limitations imply that the results should be interpreted as descriptive and relational patterns rather than precise estimates of behavioural responses or causal effects.

9 Discussion and Conclusion

Summary of findings This paper provides a comprehensive empirical analysis of the relationship between climate conditions and tourism activity across European regions. The analysis is based on a harmonised panel dataset covering 273 NUTS2 regions across 34 European countries over the period 2010–2023, combining tourism, economic, and climate data within a unified empirical framework.

The results show that climate variables are significantly associated with tourism, but their explanatory power remains limited once regional heterogeneity is accounted for.

More precisely, the analysis shows that temperature has a positive and statistically significant within-region effect. In the fixed effects specification, a one-degree increase relative to usual regional conditions is associated with an increase of approximately 17% in tourism activity. In contrast, precipitation does not exhibit a robust relationship once regional heterogeneity is controlled for.

Despite these effects, climate variables remain quantitatively small compared to structural economic factors. Across all specifications, GDP consistently emerges as the primary driver of tourism activity, highlighting the dominant role of economic conditions in shaping regional tourism patterns.

The analysis also highlights the presence of non-linear patterns, although their form depends on model specification. While the econometric approach suggests an inverted U-shaped relationship, the machine learning results indicate a more gradual and predominantly increasing pattern. This suggests that climate effects are present but relatively weak and sensitive to modelling assumptions.

Overall, these findings suggest that climate influences tourism at the margin, but does not fundamentally reshape tourism patterns over the period considered. The results therefore provide robust evidence, based on a large-scale comparative framework, that structural economic differences remain the central drivers of tourism dynamics across European regions.

Interpretation and implications Taken together, these results suggest that climate influences tourism primarily at the margin rather than as a central determinant. Short-term temperature deviations affect tourism activity within regions, but these effects remain limited relative to persistent economic and structural differences across regions.

The presence of non-linear effects in the econometric specification further indicates that the relationship between climate and tourism is not uniform, and may depend on local conditions and baseline climate levels. However, the absence of a clear reversal in the machine learning results suggests that such threshold effects are not strong enough to systematically reshape tourism patterns. This result remains intuitively plausible, as extremely high temperatures may reduce the attractiveness of certain destinations. However, the empirical evidence provided here does not offer a sufficiently strong or consistent pattern to draw a definitive conclusion on the existence of a clear optimal temperature threshold.

Relation to the research question More directly in relation to the broader question motivating this study, the results provide only partial support for the hypothesis that tourism demand may shift toward cooler regions in a context of rising temperatures. While the econometric analysis suggests the existence of an optimal temperature range, the overall evidence does not indicate a clear decline in tourism at higher temperature levels.

As such, the findings do not provide strong evidence of a systematic redistribution of tourism toward cooler destinations within the observed period. Instead, they suggest that any potential shift is likely to be gradual and mediated by regional characteristics rather than driven solely by climatic conditions.

Methodological contribution From a methodological perspective, this study illustrates both the potential and the limits of data-driven approaches in computational social science. The combination of econometric and machine learning methods provides complementary insights: econometric models offer interpretable estimates of within-region effects, while machine learning models allow for flexible exploration of non-linearities.

At the same time, the differences between these approaches highlight the importance of model choice and data structure. In particular, the dominance of cross-sectional variation in the data tends to favour structural variables such as GDP, potentially masking more subtle climate effects.

Implications for climate change and tourism More broadly, these findings suggest that while climate change may influence tourism patterns over time, its effects are likely to interact with existing economic structures rather than fundamentally reshape them in the short run. Regions with strong economic and tourism infrastructure may remain attractive despite changes in climate conditions, while less developed regions may not necessarily benefit from more favourable temperatures.

Future research Future research could extend this work by incorporating higher-frequency climate data to better capture seasonal dynamics, or by exploiting quasi-experimental settings to improve causal identification. Integrating richer contextual variables, such as infrastructure, accessibility, or tourism specialisation, would also help to better understand the mechanisms underlying the observed patterns.

More broadly, further research should investigate how climate change may influence the spatial distribution of tourism demand over the long term. Such analysis would be essential to anticipate potential changes in regional development, urbanisation patterns, and mobility systems, particularly in a context where tourism plays an increasingly important role in local economies.

Overall, this study contributes to a better understanding of climate-tourism relationships by highlighting both their existence and their limits, and by showing that structural economic factors remain central in shaping tourism dynamics across European regions.

Data and Code Availability

All data, code, and presentation materials are publicly available in the following repository:

`https://github.com/lolipop913/CSS_Climate-Tourism-Europe_HenriVasserot`

The repository includes the cleaned dataset, data processing scripts, replication code, and presentation materials used in this study.

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